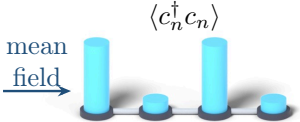
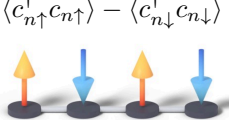


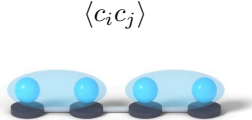
$H = \sum_{ij} t_{ij}\, c_i^\dagger c_j$
$+ \sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j^\dagger c_k c_l$
eq. (1.5)



charge density wave
breaks **translational**
eq. (1.6)
not revisited



magnet
breaks **time-reversal**
eqs. (1.7) and (1.8)
again in session 6



superconductor
breaks **gauge**
eqs. (1.9) and (1.10)
again in session 5